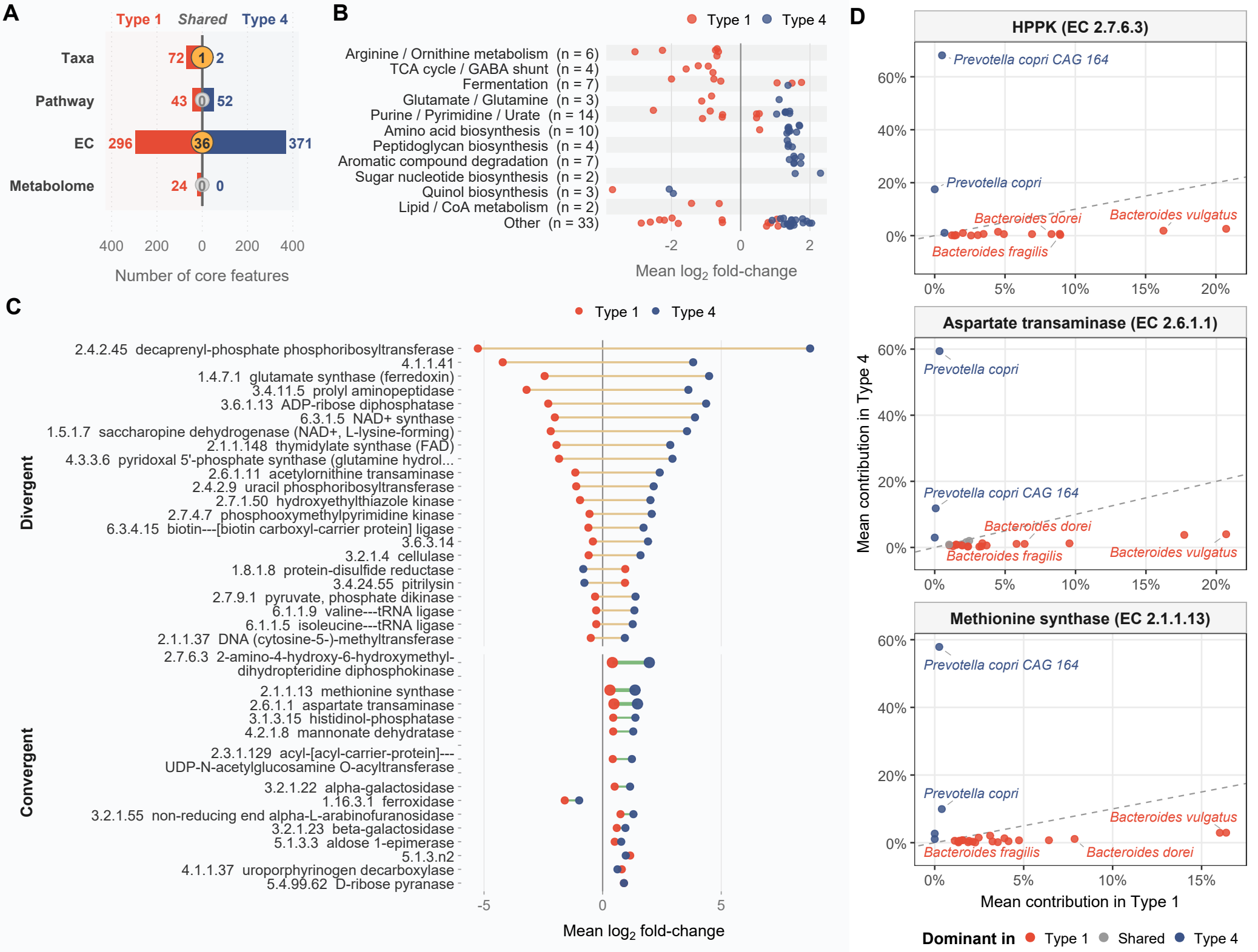

A
Type 1
Shared
Type 4
Taxa
72
1
2
Pathway
43
0
52
EC
296
36
371
Metabolome
24
0
0
400
200
0
200
400
Number of core features
B
Type 1
Type 4
Arginine / Ornithine metabolism (n = 6)
TCA cycle / GABA shunt (n = 4)
Fermentation (n = 7)
Glutamate / Glutamine (n = 3)
Purine / Pyrimidine / Urate (n = 14)
Amino acid biosynthesis (n = 10)
Peptidoglycan biosynthesis (n = 4)
Aromatic compound degradation (n = 7)
Sugar nucleotide biosynthesis (n = 2)
Quinol biosynthesis (n = 3)
Lipid / CoA metabolism (n = 2)
Other (n = 33)
-2
0
2
Mean log2 fold-change
C
Type 1
Type 4
Divergent
2.4.2.45 decaprenyl-phosphate phosphoribosyltransferase
4.1.1.41
1.4.7.1 glutamate synthase (ferredoxin)
3.4.11.5 prolyl aminopeptidase
3.6.1.13 ADP-ribose diphosphatase
6.3.1.5 NAD+ synthase
1.5.1.7 saccharopine dehydrogenase (NAD+, L-lysine-forming)
2.1.1.148 thymidylate synthase (FAD)
4.3.3.6 pyridoxal 5'-phosphate synthase (glutamine hydrol...
2.6.1.11 acetylornithine transaminase
2.4.2.9 uracil phosphoribosyltransferase
2.7.1.50 hydroxyethylthiazole kinase
2.7.4.7 phosphooxymethylpyrimidine kinase
6.3.4.15 biotin---[biotin carboxyl-carrier protein] ligase
3.6.3.14
3.2.1.4 cellulase
1.8.1.8 protein-disulfide reductase
3.4.24.55 pitrilysin
2.7.9.1 pyruvate, phosphate dikinase
6.1.1.9 valine---tRNA ligase
6.1.1.5 isoleucine---tRNA ligase
2.1.1.37 DNA (cytosine-5-)-methyltransferase
Convergent
2.7.6.3 2-amino-4-hydroxy-6-hydroxymethyl-dihydropteridine diphosphokinase
2.1.1.13 methionine synthase
2.6.1.1 aspartate transaminase
3.1.3.15 histidinol-phosphatase
4.2.1.8 mannonate dehydratase
2.3.1.129 acyl-[acyl-carrier-protein]---UDP-N-acetylglucosamine O-acyltransferase
3.2.1.22 alpha-galactosidase
1.16.3.1 ferroxidase
3.2.1.55 non-reducing end alpha-L-arabinofuranosidase
3.2.1.23 beta-galactosidase
5.1.3.3 aldose 1-epimerase
5.1.3.n2
4.1.1.37 uroporphyrinogen decarboxylase
5.4.99.62 D-ribose pyranase
-5
0
5
Mean log2 fold-change
D
HPPK (EC 2.7.6.3)
Prevotella copri CAG 164
Prevotella copri
Bacteroides dorei
Bacteroides vulgatus
Bacteroides fragilis
Aspartate transaminase (EC 2.6.1.1)
Prevotella copri
Prevotella copri CAG 164
Bacteroides dorei
Bacteroides fragilis
Bacteroides vulgatus
Methionine synthase (EC 2.1.1.13)
Prevotella copri CAG 164
Prevotella copri
Bacteroides vulgatus
Bacteroides fragilis
Bacteroides dorei
0%
5%
10%
15%
20%
40%
60%
Mean contribution in Type 4
Mean contribution in Type 1
Dominant in
Type 1
Shared
Type 4